
When Shift Happens - Confounding Is to Blame

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Distribution shifts alter the data-generating distribution and thereby degrade the
2 robustness and generalization properties of machine learning predictors. Although a
3 standard causal perspective suggests that robustness should be improved by learning
4 causal-invariant representations, recent empirical evidence yields two technically
5 important observations: (i) empirical risk minimization (ERM) can match or
6 exceed state-of-the-art out-of-distribution (OOD) generalization methods, and (ii)
7 ERM attains stronger OOD performance when trained on all available covariates
8 rather than only on variables judged to be causal. We use empirical evidence and
9 information-theoretic analysis to attribute this behavior to hidden confounding.
10 Shifts in hidden confounders perturb the observed joint distribution in ways that
11 violate assumptions imposed by many existing OOD generalization objectives.
12 Under these perturbations, we prove that effective prediction may require estimating
13 environment-specific input–output relationships rather than enforcing only invariant
14 relationships. We further show that augmenting predictors with proxy variables for
15 hidden confounders can reduce the adverse effects of hidden confounding shifts.
16 The analysis provides formal guidance for OOD algorithm design and for covariate
17 selection under unobserved confounding.

1 Introduction

18 Generalization denotes the capacity to draw reliable conclusions for unobserved samples from
19 information available in observed samples, and this requirement appears across multiple technical
20 domains. In medicine and the social sciences, it is formalized through *external validity*, where
21 an effect or conclusion estimated in one population must transfer to another population [13]; in
22 ecology, it motivates *space-for-time substitutions*, where spatial variation is used as an empirical
23 surrogate for temporal change in long-horizon ecological inference [60]; and in engineering, it
24 underlies *robust control*, where a controller must satisfy performance specifications under unmodeled
25 disturbances [35]. Modern machine learning provides a broad class of estimation procedures for
26 constructing such generalizable predictive models [100, 88, 49].

28 For a learned predictor to generalize under deployment, its risk must remain controlled when the
29 test distribution differs from the training distribution. For example, a model trained to predict *food*
30 *stamp reciprocity* from *household attributes* in one state should retain predictive accuracy when the
31 same prediction task is instantiated in another state. We say that a model achieves *out-of-distribution*
32 (*OOD*) *generalization* when its performance remains stable on both in-distribution (ID) data, from
33 which the training samples are drawn, and OOD test data, whose distribution may differ. Over the last
34 decade, domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81],
35 and robust learning [42, 69, 94] have developed objectives with provable guarantees under specified
36 distributional assumptions. Nevertheless, when model selection is carefully controlled [30, 38, 87],
37 predictors trained by standard *empirical risk minimization (ERM)* [86] often achieve competitive
38 OOD performance in real-world applications [30, 24, 50, 57, 67, 87]. In addition, a recent empirical
39 analysis of 16 real-world tabular datasets reports that using all observed covariates, rather than
40 restricting the model to variables with direct causal relevance to the outcome, can improve OOD
41 generalization [57]. These observations are inconsistent with a purely invariance-based explanation
42 and require a more explicit account of the mechanisms that generate OOD performance.

Distribution shifts typically arise when observations are sampled from multiple environments [71]. For instance, *food stamp reciprocity* can vary by state because eligibility rules and administrative procedures are state-specific, yielding environment-indexed distributions. In such settings, some statistical relationships between covariates and the outcome may be stable across environments; these relationships are referred to as *invariant relationships* [6]. Many OOD methods seek to identify and exploit such invariances to obtain predictors with stable test performance [6, 59, 31, 66, 56, 98, 63]. A particularly important subclass is *causal invariance*, in which the causal mechanisms in the data-generating process remain unchanged across environments [71, 59, 6]. In real-world data, however, not all covariates relevant for predicting the outcome are observed, due to collection constraints, privacy restrictions, or measurement error [14, 52, 20, 48]. Missing variables can invalidate the observed invariances on which these methods rely, especially when the missing variables are confounders that jointly affect observed covariates and the outcome. Such hidden confounders are common, and environment-dependent changes in their distributions define hidden confounding shifts. If these shifts are ignored, a predictor may incur degraded OOD performance [40, 4, 85, 62] and may also estimate a statistically incorrect covariate–outcome relationship.

Despite its practical importance, the effect of hidden confounding shifts on OOD generalization has received limited formal treatment [4, 40, 85, 62]. This work develops theoretical and empirical explanations for recent benchmark findings on ERM, covariate inclusion, and OOD performance [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- We formulate the role of confounding shift in OOD generalization from a causal perspective (§ 3) and specify why informative, non-causal covariates can improve predictive performance.
- We show that maximizing predictive information between model predictions and true outcomes requires explicit estimation of environment-specific relationships; invariant representations alone, while necessary, are not sufficient (§ 4.1, § 4.2).
- We demonstrate that variables informative about either the outcome or the hidden confounders increase predictive information between model predictions and true outcomes, which establishes the importance of covariate selection under hidden confounding shift (§ 4.3).
- Experiments on real-world and synthetic datasets show that (i) hidden confounding is prevalent in real-world tabular benchmarks, (ii) estimation of environment-specific relationships is positively associated with ID and OOD test accuracy, and (iii) inclusion of informative, non-causal covariates improves generalization (§ 5).

Overall, the analysis explains why ERM can remain effective under real-world distribution shifts and identifies hidden confounding as a central structural source of OOD difficulty that must be represented explicitly in future methods.

2 Related work

Out-of-distribution (OOD) generalization includes several related problem formulations, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. Domain adaptation assumes unlabeled samples from the target distribution, whereas domain generalization uses environment labels only during training. Federated learning specifies a distributed optimization setting in which communication and privacy constraints interact with multi-environment data. OOD detection instead treats distribution shift as a sample-level testing problem. Many methods in these families impose an invariance criterion on learned relationships, but recent work shows that invariance alone does not specify a sufficient inductive bias for generalization [46, 72, 93]. In particular, Ye et al. [93] argue that feature invariance must be coupled with feature *informativeness*. Our analysis therefore treats invariance and informativeness as distinct information-theoretic quantities that jointly determine OOD behavior.

Proxy variable adjustment: When confounders are observed during training but unobserved at test time, Landeiro and Culotta [40] propose an adjustment procedure that accounts for confounding shifts in order to improve classifier performance. Building on proxy-adjustment methods for causal effect estimation [55, 39], recent domain adaptation methods formulate unknown distribution shift as a causal effect identification problem [4, 85]. OOD generalization under hidden confounding shift has also been studied under an overlapping-confounder-support assumption [62]. Our contribution differs in that we characterize *how* proxy variables affect the information terms that control OOD generalization.

All variable models vs causal models. Recently, Nastl and Hardt [57] introduced a benchmark in which covariates are partitioned into causal, arguably causal, anti-causal, and other spurious groups. Across 16 benchmark datasets, they find that predictors using all covariates Pareto-dominate predictors restricted to causal or arguably causal subsets on both ID and OOD data. Existing theory provides only limited explanation for this empirical pattern, and we supply settings and formal arguments that account for it. For linear causal models, anchor regression [68] provides a framework that interpolates between using all observed covariates and using only causal covariates. Our objective is to specify the effect of adding covariates that are not necessarily causal when the data are subject to hidden confounding shift. Eastwood et al. [21] show that unstable covariates can improve performance when they contain label information and are conditionally independent of stable covariates given the label; their procedure adjusts for distribution shift using unlabeled test-domain data. However, on a real medical dataset not constructed for this problem [8], ERM remains competitive with their method, consistent with Nastl and Hardt [57]. This observation is also compatible with the result that maximum likelihood estimation (MLE) is minimax optimal for OOD generalization under well-specified covariate shifts [26]. Real-world data, however, are often misspecified by unobserved confounding shift, which is the central setting studied here.

3 Manifestations of hidden confounding shift

This section formalizes hidden confounding shift and motivates its relevance through a causal-effect-identification example involving observed covariates and the outcome.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , the observed distributions may differ between environments e and e' in the forms listed in Table 1. We treat these shifts as consequences of changes in the distribution $\mathbb{P}(U)$ of an unobserved variable U that causally affects $\mathbf{X}$, Y , or both (Figure 1). A perturbation in $\mathbb{P}(U)$ is then transmitted to observed distributions over $\mathbf{X}$ and Y . For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U) \mathbb{P}(\mathbf{X} | U)$, so if $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, then $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same marginalization argument applies to shifts in $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

Many OOD generalization methods impose causal restrictions on U , $\mathbf{X}$, Y that imply particular invariances. If $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), then $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y) \mathbb{P}(\mathbf{X} | Y)$, and a change in $\mathbb{P}(Y)$ produces label shift, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, while preserving the conditional covariate distribution $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. If $U \rightarrow \mathbf{X} \rightarrow Y$ instead (Figure 1 (b)), then $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$ induces covariate shift while the conditional law $\mathbb{P}(Y | \mathbf{X})$ remains invariant, i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These invariances need not hold when U jointly causes $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In that case, an environment-level shift in $\mathbb{P}(U)$ can simultaneously induce label shift, covariate shift, conditional covariate shift, and concept shift. The corresponding invariant object is instead $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, known as *stable confounding shift* [85, 4]; because U is unobserved, this invariance is not directly available in the observed distribution.

Information-theoretic measures: Following Federici et al. [22], we quantify distribution shift with mutual information. Let E denote the environment variable. We use $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E | Y)$, $I(Y; E | \mathbf{X})$, $I(\mathbf{X}, Y; E)$ to measure environment dependence in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, respectively. Standard OOD formulations often impose either $I(\mathbf{X}; E | Y) = 0$ via a label-shift assumption or $I(Y; E | \mathbf{X}) = 0$ via a covariate-shift assumption. Under hidden confounding shift, both conditional mutual-information terms can be nonzero.

Challenges with hidden confounding shift: Figure 2 gives a concrete OOD regression example in which the outcome Y depends on the observed covariate X and on an unobserved confounder U .

Table 1: Summary of distribution-shift types and their corresponding distributional inequalities.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

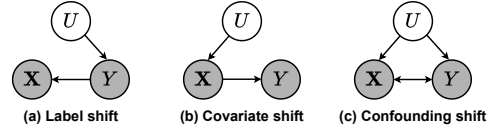


Figure 1: Causal graph structures that induce the distribution-shift regimes considered in the text.

If $U \rightarrow \mathbf{X} \rightarrow Y$ instead (Figure 1 (b)), then $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$ induces covariate shift while the conditional law $\mathbb{P}(Y | \mathbf{X})$ remains invariant, i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These invariances need not hold when U jointly causes $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In that case, an environment-level shift in $\mathbb{P}(U)$ can simultaneously induce label shift, covariate shift, conditional covariate shift, and concept shift. The corresponding invariant object is instead $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, known as *stable confounding shift* [85, 4]; because U is unobserved, this invariance is not directly available in the observed distribution.

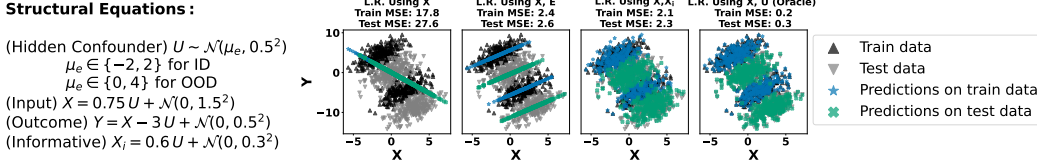


Figure 2: Evaluation of four linear-regression (L.R.) models in an OOD setting with hidden confounding shifts and limited environment overlap (distant μ_e). (i) Training only on X yields an incorrect association with Y , illustrating Simpson’s paradox. (ii) Environment-specific summary statistics of X , denoted by E , recover the correct association but have limited representational capacity. (iii) An informative covariate X_i for U improves OOD generalization. (iv) The oracle model observes both X and U .

Distinct environments are generated by environment-specific shifts in the mean of U , denoted by μ_e . A linear regression model trained only on X estimates an incorrect X – Y association, as shown in the first subplot. This behavior instantiates Simpson’s paradox [73]: within each environment, Y increases with X , whereas the pooled model recovers the opposite trend. Similar failures are observed for OOD-oriented objectives such as IRM [6], VREX [38], and Group DRO [69]. In § 4, we show theoretically that the optimal generalization strategy in this setting must estimate environment-specific relationships, which permits recovery of the correct X – Y relationship. The second subplot illustrates this mechanism: environment-specific summary statistics of X , including means, standard deviations, and quantiles, provide proxy information that partially adjusts for the confounder, analogous to the *backdoor adjustment* criterion in causal effect estimation [58]. Because summaries constructed from observed covariates have limited representational capacity, they may be insufficient by themselves. As shown in § 4.3, additional informative covariates that proxy for the unobserved confounder can further improve OOD performance, as illustrated in the third subplot. We include an oracle model trained on X and U as the best-achievable mean-squared-error reference.

4 Impact of hidden confounding shifts on OOD generalization

This section gives an information-theoretic analysis of how hidden confounding shifts affect OOD generalization objectives. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the predicted label for input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation. We write $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ for entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ for mutual information. The environment-specific risk of $(f \circ \phi)$ in environment e is $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X},Y) \sim \mathbb{P}_{\mathbf{X},Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$, where $\ell(\cdot, \cdot)$ is the loss. The OOD objective is to learn $(f \circ \phi)$ with controlled risk on both ID and OOD distributions. Robust optimization methods [7, 9, 19, 75, 69] minimize the worst training-environment risk, $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, where $\mathcal{E}_{tr}$ is the set of training environments. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, with hyperparameter β . Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to f being simultaneously optimal across all environments. Empirical risk minimization (ERM) pools all training environments and minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X},Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ denotes the observed data in environment e [6, 38].

To relate these objectives to hidden confounding shift, we analyze predictive information $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures the information that the prediction $\hat{Y}$ retains about the true label Y and therefore provides a task-agnostic criterion for prediction quality. We first define two representation properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that are central to OOD generalization [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness quantifies the information that $\phi(\mathbf{X})$ provides about Y after conditioning on the environment. Decreasing CINF discards label-relevant information, whereas increasing CINF can be undesirable in fairness-sensitive settings because $\phi(\mathbf{X})$ may encode environment-specific sensitive or biased signals. Such representations can be discouraged by minimizing $I(\phi(\mathbf{X}); E)$, but this constraint may reduce predictive performance, as shown in § 4.2. In the setting considered here,

where E is not assumed to encode sensitive information, we analyze the regime in which increasing CINP can improve generalization, as formalized in § 4.1.

Definition 4.2 (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$*

Invariance measures the stability of $\phi(\mathbf{X})$ across environments after conditioning on the label Y . Reducing invariance preserves environment-specific information in the representation and can induce overfitting to training environments; increasing invariance suppresses such dependencies and implements invariant learning. Extending the information-theoretic shift measures from § 3, we use $I(\phi(\mathbf{X}); E)$ for *feature shift* and $I(Y; E | \phi(\mathbf{X}))$ for *concept shift*. Depending on whether the representation or the raw covariates are under discussion, *concept shift* denotes either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

4.1 A general decomposition of predictive information

We begin by specifying the causal-graph terminology used in the decomposition. A causal graph $\mathcal{G}$ contains nodes representing random variables and directed edges representing direct causal effects. A *path* between X_i and X_j is a sequence of distinct nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is a path whose arrows all point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. Along such a path, X_i is a *parent* of X_{i+1} , X_{i+1} is a *child* of X_i , X_i is an *ancestor* of X_j , and X_j is a *descendant* of X_i . Any path can be decomposed into chains $X_i \rightarrow X_j \rightarrow X_k$, forks $X_i \leftarrow X_j \rightarrow X_k$, and colliders $X_i \rightarrow X_j \leftarrow X_k$. In a chain or fork, X_i and X_k are marginally dependent and become conditionally independent after conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent and become conditionally dependent after conditioning on X_j or one of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is *blocked* by a conditioning set $\mathcal{S}$ if either (i) the path contains a chain or fork whose middle node is in $\mathcal{S}$, or (ii) the path contains a collider such that neither the collider nor any descendant of the collider is in $\mathcal{S}$. If $\mathcal{S}$ blocks all paths from X_i to X_k , then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$; otherwise, the path remains open.

Consider predictive information $I(Y; \hat{Y})$, where $\hat{Y} = (f \circ \phi)(\mathbf{X})$ is computed from the learned representation $\phi(\mathbf{X})$. Figure 3 specifies the causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X})$, $\hat{Y}$, and E , with U denoting a hidden confounder. Conditional on either $\mathbf{X}$ or $\phi(\mathbf{X})$, the prediction $\hat{Y}$ contains no additional information about Y ; equivalently, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. The environment variable E indexes shifts in $\mathbb{P}(U)$, so distinct environments correspond to different confounder distributions. We first derive a general decomposition of predictive information without specifying how U causally affects $\mathbf{X}$ and Y . The explicit treatment of U 's influence appears in § 4.2.

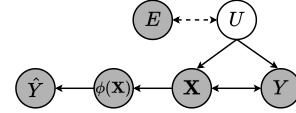


Figure 3: The bidirected relation between $\mathbf{X}$ and Y denotes that some covariates in $\mathbf{X}$ may cause Y , while other covariates may be caused by Y .

Proposition 4.1. *For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \frac{\overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}}}{2} + \frac{\overbrace{I(Y; E)}^{\text{Label shift}}}{2} + \frac{\overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}}}{2} - \frac{\overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}}}{2} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are given in Appendix § A. Equation (1) provides a common information-theoretic coordinate system for existing OOD objectives, which can be interpreted as optimizing different terms in the decomposition. IRM [6], for example, constrains a fixed classifier f to be optimal in all environments; this constraint is aligned with maximizing $-I(\phi(\mathbf{X}); E|Y)$, because additional environment-specific information in $\phi(\mathbf{X})$ conditional on Y would generally make a single classifier non-optimal across

environments. DANN [23] and the independence criterion for fair classification [22] suppress environment information by minimizing $I(\phi(\mathbf{X}); E)$. Domain-adaptation methods likewise enforce domain-invariant feature properties [82, 90]. Although minimizing $I(\phi(\mathbf{X}); E)$ can be appropriate for fair classification, Equation (1) shows that it can also reduce $I(Y; \hat{Y})$ and thereby harm predictive performance [98, 33, 22]. CDAN [51] instead targets joint alignment, $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. This equality makes E independent of $(\phi(\mathbf{X}), Y)$ and implies $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, joint-distribution alignment drives this conditional mutual information toward zero. The term $I(Y; E)$ is constant for a fixed dataset, and separate robust-learning methods address shifts in label distributions [69, 95].

Because Equation (1) does not specify how U affects $\mathbf{X}$ and Y , causal structure determines which terms should be controlled. If $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, making minimization of $I(\phi(\mathbf{X}); E | Y)$ a principled objective. If $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, which motivates minimizing $I(Y; E | \phi(\mathbf{X}))$. When $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, however, the terms in Equation (1) interact through hidden confounding, and the appropriate optimization strategy is no longer immediate. The next section resolves this case.

4.2 Predictive information under hidden confounding shift

To characterize the effect of a hidden confounder U that causes both $\mathbf{X}$ and Y , we analyze *two special cases* for the observed covariate–label relationship: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider Y , thereby allowing additional information flow from $\phi(\mathbf{X})$ to E through Y . By contrast, conditioning on $\phi(\mathbf{X})$ partially blocks information flow from Y to E at $\mathbf{X}$ whenever $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

When $Y \rightarrow \mathbf{X}$, the inequalities in (3) follow from the complementary graphical relations. Conditioning on Y blocks information flow from $\phi(\mathbf{X})$ to E at Y , whereas conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E through the collider $\mathbf{X}$, because $\phi(\mathbf{X})$ is a child of $\mathbf{X}$ and conditioning on a descendant of a collider opens the collider path [58] (see the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E | Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E | \phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Using the inequalities in (2) and (3), we obtain the following specialization of the predictive-information decomposition.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}^{\text{Residual}} \quad (4)$$

Proposition 4.2 treats $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately. The former structure is more common in real-world tabular prediction: 11 of the 16 datasets considered in recent benchmark studies [57, 50, 24] follow $\mathbf{X} \rightarrow Y$. In the remaining 5 datasets, the number of covariates caused by Y is substantially smaller than the number of covariates that cause Y . The following statement summarizes the implication of Proposition 4.2.

Relative to (1), the specialized decomposition in (4) for (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ eliminates the explicit *variation, label shift, feature shift, and concept shift* terms. Therefore, under hidden confounding shift with either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, maximizing predictive information reduces to maximizing the contrast *conditional informativeness* – *residual*.

Maximizing conditional informativeness: Recent work by Prashant et al. [62] proposes a mixture-of-experts (MoE) architecture for OOD generalization under hidden confounding shift, where each

expert corresponds to a particular hidden-confounder assignment. Each expert therefore optimizes performance within the environment associated with that confounder value. This architecture is aligned with maximizing conditional informativeness $I(\phi(\mathbf{X}); Y|E)$, and Equation (4) gives a theoretical justification for MoE-type approaches. At the same time, Equation (4) indicates the need for algorithms that apply beyond the assumptions of that model. In particular, Prashant et al. [62] assume overlapping confounder support, $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and discrete-valued proxy variables.

Figure 2 shows that, even without confounder-support overlap and with a continuous-valued proxy, linear regression can recover the correct causal relationship between observed covariates and the target when environment-specific information is supplied. This example motivates OOD generalization methods that remain valid when confounder support is limited and the confounder or its proxies are continuous-valued.

291

Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ measures label information contained in $\phi(\mathbf{X})$ that is not transmitted to the prediction $\hat{Y}$. It can be written as $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is related to expected cross-entropy through $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29]. The term $H(Y | \hat{Y})$ is the *refinement error*, which quantifies the ability of the model scores to separate classes. Minimizing $I(\phi(\mathbf{X}); Y | \hat{Y})$ can therefore be approached in two complementary ways. One can directly enforce conditional independence between the representation and the label given the prediction, $Y \perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can reduce refinement error $H(Y | \hat{Y})$, for example by using training procedures that improve predictive refinement before optimizing calibration error [10].

301 4.3 Impact of additional informative variables

Section 4.2 characterizes the optimal information terms when the confounder itself is unobserved. In many applications, however, observed variables may serve as proxies for the hidden confounder [4, 85, 62]. We therefore analyze how adding non-causal but informative covariates changes predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp \mathbf{X}_I | \mathbf{X}, E$.*

Because $U \rightarrow Y$, a covariate informative about U is also informative about Y , and conversely in this setting. Equation (1) indicates that maximizing $I(Y; \hat{Y})$ requires reducing concept shift $I(Y; E | \phi(\mathbf{X}))$. Based on empirical evidence, Liu et al. [50] suggest that collecting additional covariates $\mathbf{X}_I$ that make $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ more stable across environments, and thereby reduce concept shift, improves OOD test accuracy. We prove that adding informative variables changes the decomposition in directions that can increase predictive information.

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) > I(\phi_1(\mathbf{X}); Y | E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E | \phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

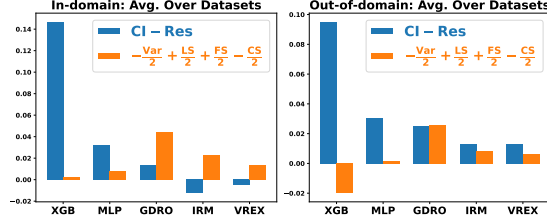
Proposition 4.3 states that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii), while decreasing concept shift (5.iii). The same operation also increases variation (5.iv), so practical algorithms may require explicit mechanisms for controlling this term.

320 5 Experimental results

We evaluate the theory on real-world and synthetic datasets with three goals: (i) detecting empirical signatures of hidden confounding shift in real-world data, (ii) measuring how the terms in (1) relate to OOD generalization under hidden confounding shift, and (iii) quantifying the effect of informative covariates on generalization. The real-world evaluation uses eight tabular benchmark datasets, *Food stamps, Readmission, Income, Public coverage, Unemployment, Diabetes, Hypertension, and*

Table 2: Estimated distribution-shift measures. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. One-sample t-tests against zero ($\alpha = 0.05$) indicate that all reported measures are significantly nonzero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The plotted contrast *conditional informativeness* – *residual* is positively associated with average ID and OOD test accuracy over the eight datasets summarized in the table.

326 *ASSISTments*, with domain splits taken from the *TableShift* benchmark [24]. We report ID-test and
327 OOD-test accuracies. The model suite contains two ERM-based methods, XGBoost (XGB) [15]
328 and a multi-layer perceptron (MLP); two domain-generalization methods, IRM [6] and VREX [38];
329 and one robust-learning method, Group DRO (GDRO) [69]. Unless otherwise stated, results are
330 averaged over 5 random hyperparameter configurations as specified in Appendix B; seed effects are
331 statistically negligible for these datasets [24]. Additional setup details appear in Appendix § B, and
332 hyperparameter-tuned results appear in Appendix C.

333 **Hidden confounding shift:** Hidden confounding shifts imply observable changes in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$,
334 $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. Table 2 shows nonzero label shift, covariate shift, conditional covariate shift,
335 and concept shift across the real-world datasets, which is consistent with the presence of hidden
336 confounding shift. For qualitative support, we query GPT-4o [1] for plausible hidden confounders in
337 several benchmark datasets [24, 57]; the resulting analysis is reported in Appendix § E. For example,
338 in the *food stamps* dataset, unmeasured state-specific *economic policies* may affect both *household*
339 *income* and *food stamp reciprocity*.

340 **Conditional informativeness vs. accuracy:** Proposition 4.2 implies that, under hidden confounding
341 shift, predictive information is maximized by increasing the contrast *conditional informativeness* –
342 *residual*. Figure 4 reports this quantity for the first six datasets in Table 2, which are the datasets with
343 multiple training environments [57, 24]. Dataset-level results are reported in Appendix C. The contrast
344 *conditional informativeness* – *residual* is positively associated with both ID and OOD accuracy,
345 and XGB obtains both the highest value of this contrast and the strongest average performance.
346 The dataset-wise plots in Appendix C further show that the orange components are often close to
347 zero, consistent with Proposition 4.2, because $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 -$
348 $\text{concept shift} / 2$ cancels under hidden confounding shift. The relative ID/OOD accuracies are also
349 consistent with prior reports [57, 24].

350 **Informative covariates vs. accuracy:** We next evaluate whether adding informative covariates
351 improves OOD generalization. We use the covariate partition from [57], in which available vari-
352 ables form three nested sets: *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates*
353 (Appendix § B). For each transition from a smaller to a larger covariate set, we measure how the
354 terms in (1) change jointly with ID and OOD accuracy. We summarize this relationship using the *sign*
355 *consistency metric*. For terms with positive coefficients in (1), such as conditional informativeness,
356 the metric counts how often the term increases when accuracy increases. For terms with negative
357 coefficients, such as concept shift, it counts how often the term decreases when accuracy increases.

Table 3: Sign consistency is high for concept shift (CS) across all methods. Proposition 4.2 identifies conditional informativeness (CI) as critical for generalization, and XGB has the strongest consistency on this term. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

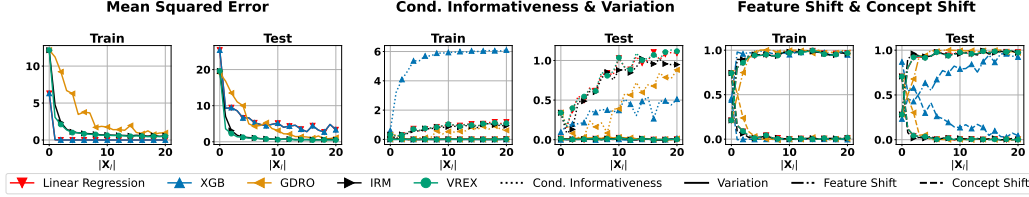


Figure 5: Increasing the number of proxy variables $\mathbf{X}_I$ for U that are informative about Y reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

Thus, the metric estimates the fraction of covariate-set transitions in which a term moves in its beneficial direction; the formal definition is provided in Appendix § B. Table 3 shows high sign consistency for both conditional informativeness and concept shift. Although reducing concept shift is important across methods, XGB also achieves stronger increases in conditional informativeness, which corresponds to additional accuracy gains. Dataset-specific results are reported in Appendix § C.

In real-world datasets, it is generally not possible to verify that *all available covariates* contain *every relevant or sufficiently informative* variable. We therefore conduct a synthetic experiment with known causal structure $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . As Figure 5 shows, adding $\mathbf{X}_I$ increases conditional informativeness and feature shift while decreasing concept shift. The variation component remains zero in this simplified setting. Synthetic data generation details and additional results are provided in Appendix § D.

6 Discussion, limitations, and future work

In summary, we decompose predictive information between model outputs and true outcomes into terms that expose the mechanisms limiting OOD generalization (Equation (1)). The analysis shows that hidden confounding shifts, expressed as environment-dependent changes in covariate and outcome distributions, create ambiguity that cannot in general be eliminated by any finite set of observed covariates or proxies. Because confounding couples covariate, label, concept, and feature shifts (inequalities in (2), (3)), methods that optimize these shifts independently are structurally incomplete. We show instead that learning environment-specific input–output mappings can be preferable to enforcing a single invariant predictor (Equation (4)). The resulting research agenda includes: (i) algorithms that remain valid under limited confounder-overlap support, (ii) formal cost–accuracy analyses for acquiring informative covariates, (iii) methods for entangled shifts that avoid untestable proxy assumptions, and (iv) uncertainty models that represent ambiguity induced by unobserved confounders.

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661 Appendix

662 A Proofs of propositions

663 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 664 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 665 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

666 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 667 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

668 *Proof.* From the causal graph in Figure 3, the substructure $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ implies that $\hat{Y}$
 669 contains no additional information about Y after conditioning on either $\phi(\mathbf{X})$ or $\mathbf{X}$. The path between
 670 $\hat{Y}$ and Y is blocked by the conditioning set $\{\phi(\mathbf{X})\}$ and also by $\{\mathbf{X}\}$, which yields $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$
 671 and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. We therefore expand $I(Y; \hat{Y})$ as follows.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

672 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

673 Adding Equations (6) and (7) and collecting like terms gives the stated decomposition:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

674 The analogous decomposition with $\mathbf{X}$ in place of $\phi(\mathbf{X})$ follows by using $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introducing
 675 $\mathbf{X}$ rather than $\phi(\mathbf{X})$ in the same expansion. $\square$

676 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 677 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 678 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

679 *Proof.* Using the causal graph in Figure 3, we prove the proposition by separately analyzing the cases
 680 $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

681 **Case 1:** $\mathbf{X} \rightarrow Y$. Conditioning on Y opens the collider path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between
 682 $\phi(\mathbf{X})$ and E , which gives the inequality

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

683 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 684 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 685 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

686 Substituting this inequality into the expansion in Equation (7) from Proposition 4.1 gives

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

687 Similarly, applying the second inequality to the expansion in Equation (6) gives

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

688 Thus $I(Y; \hat{Y})$ is upper- and lower-bounded by the same expression, $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

689 **Case 2:** $Y \rightarrow \mathbf{X}$. In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 690 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 691 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

692 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 693 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

694 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

695 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

696 Thus $I(Y; \hat{Y})$ is upper- and lower-bounded by the same expression, $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

697 Because the upper and lower bounds coincide in both cases, the desired equality follows: $I(Y; \hat{Y}) =$
 698 $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

699 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 700 $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

701 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
702 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
703 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
704 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

705 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
706 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
707 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
708 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
709 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

710 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

711 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

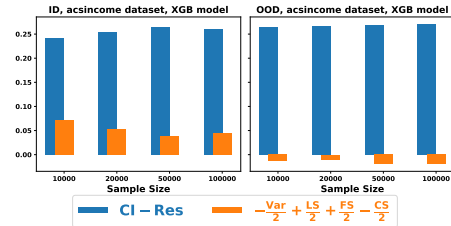
712 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
713 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
714 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
715 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
716 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

717 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
718 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
719 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
720 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
721 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

722 B Experimental setup

723 This section specifies the experimental protocol. Random-seed effects are statistically insignificant
724 in many TableShift settings [24]; therefore, the main-paper results report the mean and standard
725 deviation over five randomly sampled hyperparameter configurations from the search spaces in
726 Table B1 for each dataset–method pair. Because performance is evaluated on both ID and OOD
727 test data, we estimate the corresponding mutual-information terms with the nonparametric entropy
728 estimation toolbox [37, 77, 78]. Following [24], these estimates are computed separately on ID test
729 samples from training domains and OOD test samples from held-out domains.

730 For consistency, we sample 20,000 observations
731 from each ID and OOD test split when available;
732 datasets with fewer than 20,000 observations use
733 all available samples. Across the experiments,
734 sample size has little effect on the estimated
735 mutual-information terms. Figure B1, for exam-
736 ple, shows that the terms in Equation (1) com-
737 puted from XGB representations on the *income*
738 dataset remain stable as the sample size varies.
739 For neural models, $\phi(\mathbf{X})$ is extracted from the
740 layer immediately preceding the classification
741 head; for XGBoost, we form the representation
742 by combining output margins with SHAP values.
743 All experiments were run on a single NVIDIA RTX
744 A6000 GPU.



739 Figure B1: Estimated predictive-information terms remain stable across sample sizes on the *income* dataset.

744 For the covariate-informativeness experiments, we use the categorization of Nastl and Hardt
745 [57] and partition variables into *causal*, *arguably causal*, and *all covariates*. Causal covariates
746 have an established directional effect on the target and low plausibility of reverse causation, al-
747 though restricting to this set can omit relevant parents when domain knowledge is incomplete.
748 Arguably causal covariates have less certain causal status and satisfy at least one condition:
749 (1) known causal relevance, (2) plausible but potentially bidirectional influence, or (3) likely
750 but unconfirmed causal effect. These categories approximate causal-parent sets under available

Table B1: Hyperparameter search spaces used for the experimental evaluation.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

751 knowledge, while allowing that some relations may still be confounded. Some datasets also in-
752 clude *anti-causal* covariates for which the target likely affects the covariate rather than the re-
753 verse. The *all covariates* set contains all observed variables regardless of causal status. The
754 sign-consistency metric introduced in § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
755 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$
756 $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

757 Here, P denotes the set of ordered covariate-set pairs (s, t) such that $t \supset s$; for example, s may be the
758 causal set and t the arguably causal set. The factor of 2 accounts for evaluating both train and test splits.

759 The quantity $\mu_s^{\text{tr/te}}(m)$ is the mean value of measure m for covariate set s on training/testing data, and
760 $\mathbb{I}(\cdot)$ is the indicator function. For the three covariate settings $\{\text{causal}, \text{arguably causal}, \text{all}\}$, $|P| = 3$.
761 The final sign-consistency value for each measure averages over datasets: $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$,
762 where $|D|$ is the number of datasets.

763 C Additional results on real-world datasets

764 This section reports additional real-world results. Figures C2 and C3 compare
765 *conditional informativeness – residual* with $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 -$
766 *concept shift* / 2 for each dataset; their aggregate counterpart appears in the left panel of Fig-
767 ure 4. By Proposition 4.1, the sum of the blue and orange components is positively associated
768 with overall model performance. Under potential hidden confounding shift in real-world data,
769 however, *conditional informativeness – residual* contributes more to predictive information than
770 $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. This pattern supports objectives
771 that increase conditional informativeness under hidden confounding shift.

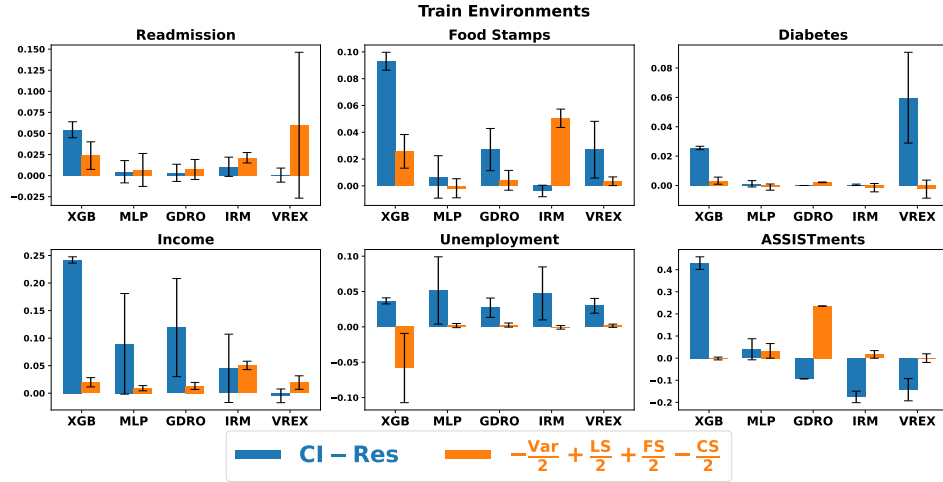


Figure C2: Dataset-specific decomposition results on ID data.

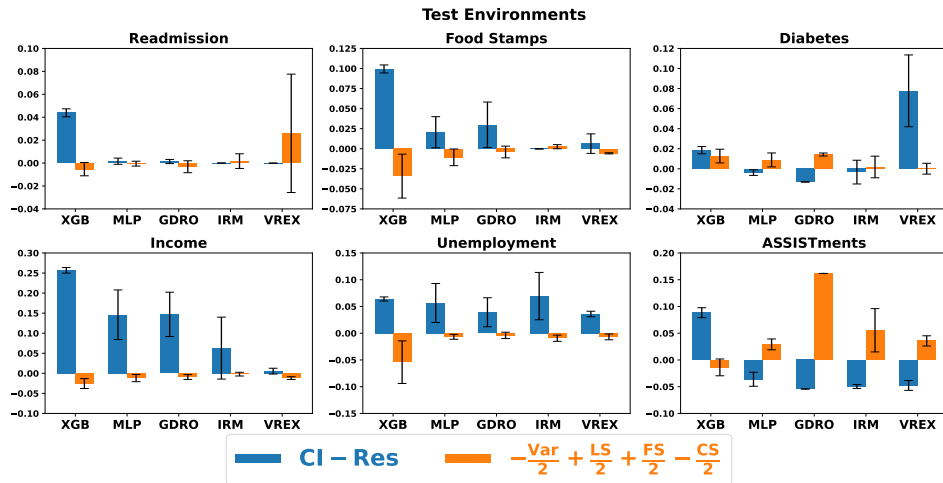


Figure C3: Dataset-specific decomposition results on OOD data.

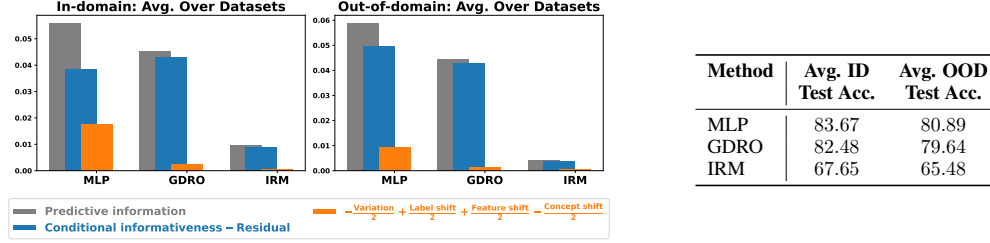


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

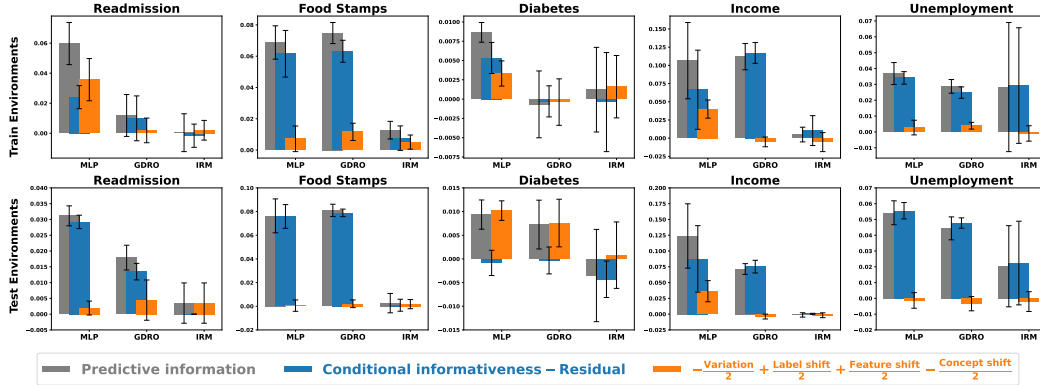


Figure C5: Dataset specific results. Comparison of the terms in predictive information.

Table C2: Comparison of ID and OOD test accuracies of models.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further show the comparison of various terms of the information decomposition in (1) for each method and dataset. From these results, we observe that the values plotted in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. Similarly, we observe that the values plotted in the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. In almost all settings, the difference *conditional informativeness* - *residual* is the majority contributor for the predictive information.

Hyperparameter tuning: We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

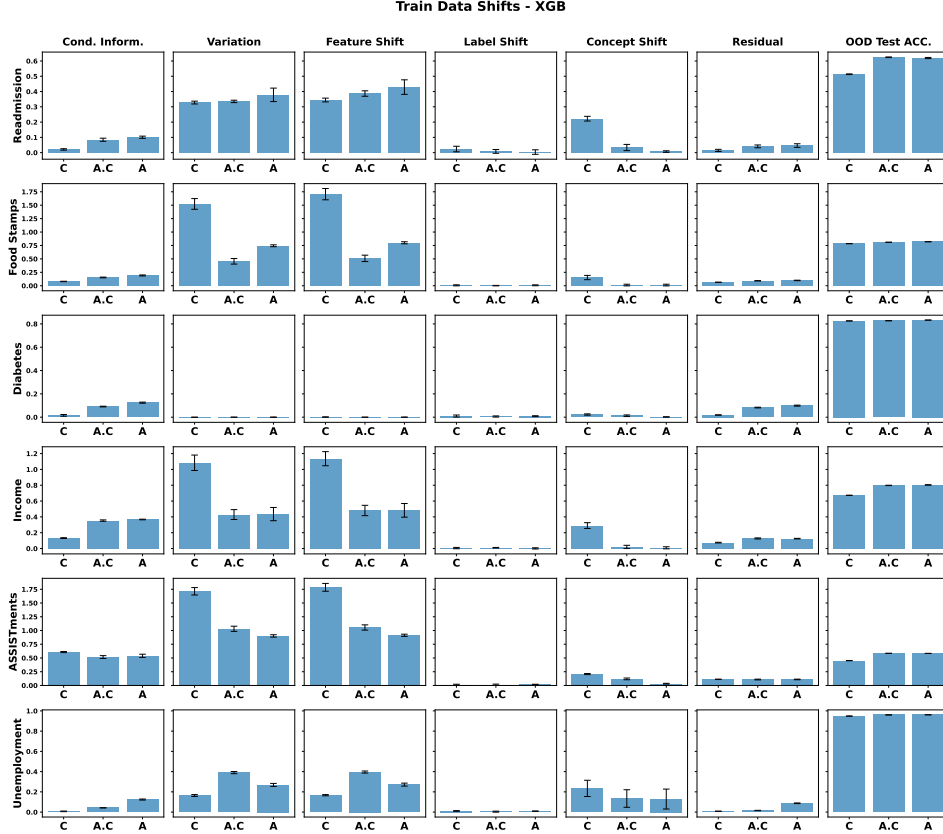


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$. Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

Potential limitations with predictive information decomposition analysis: While predictive information $I(Y; \hat{Y})$ measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

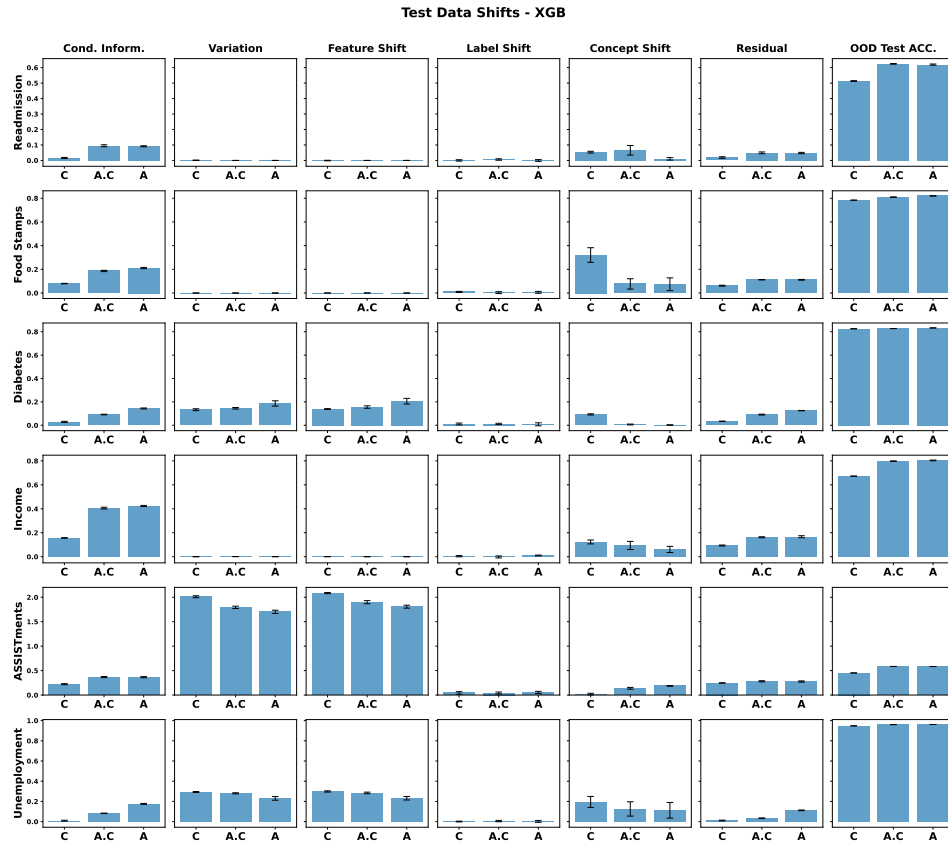


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

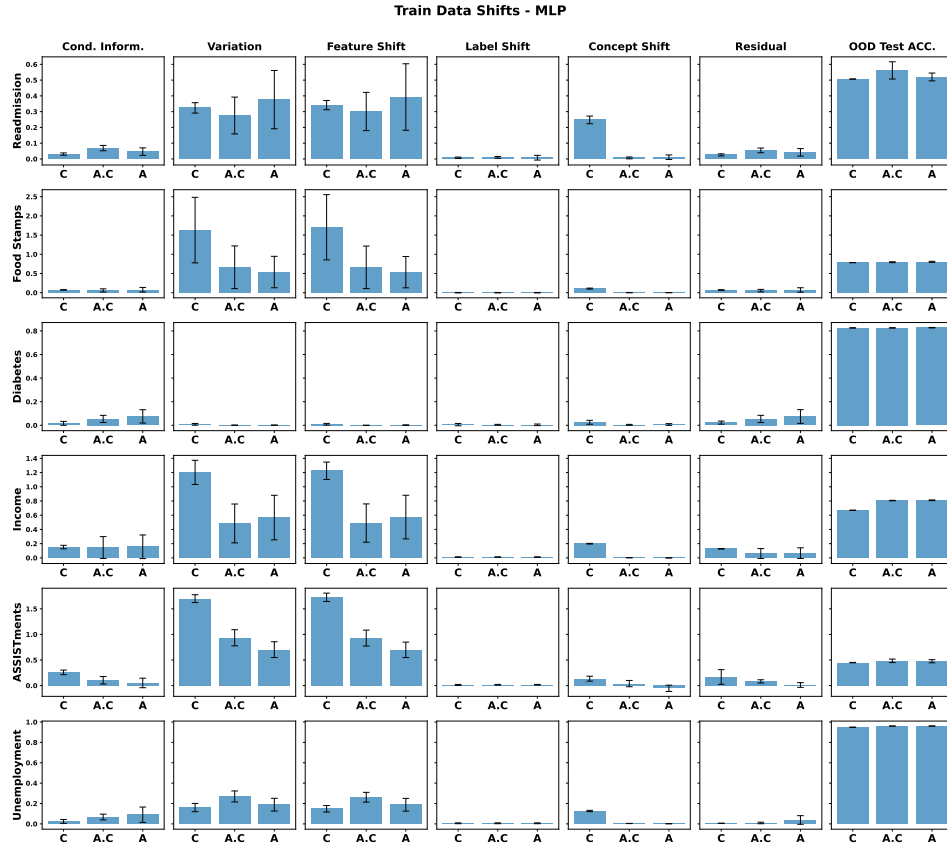


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

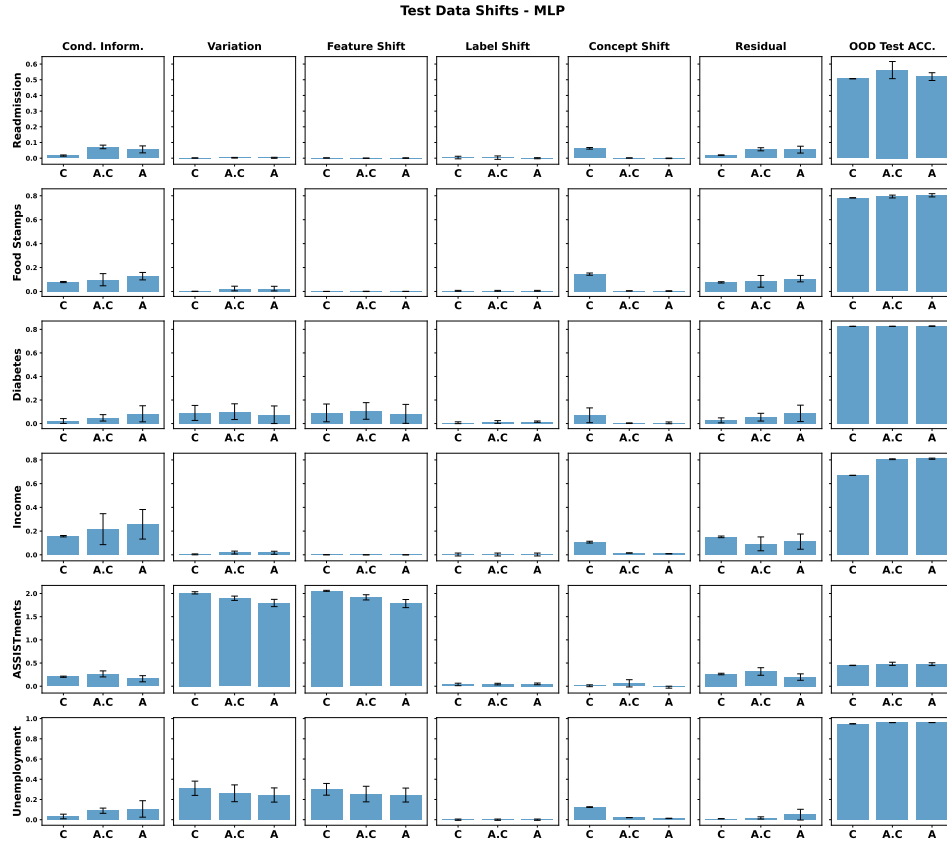


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

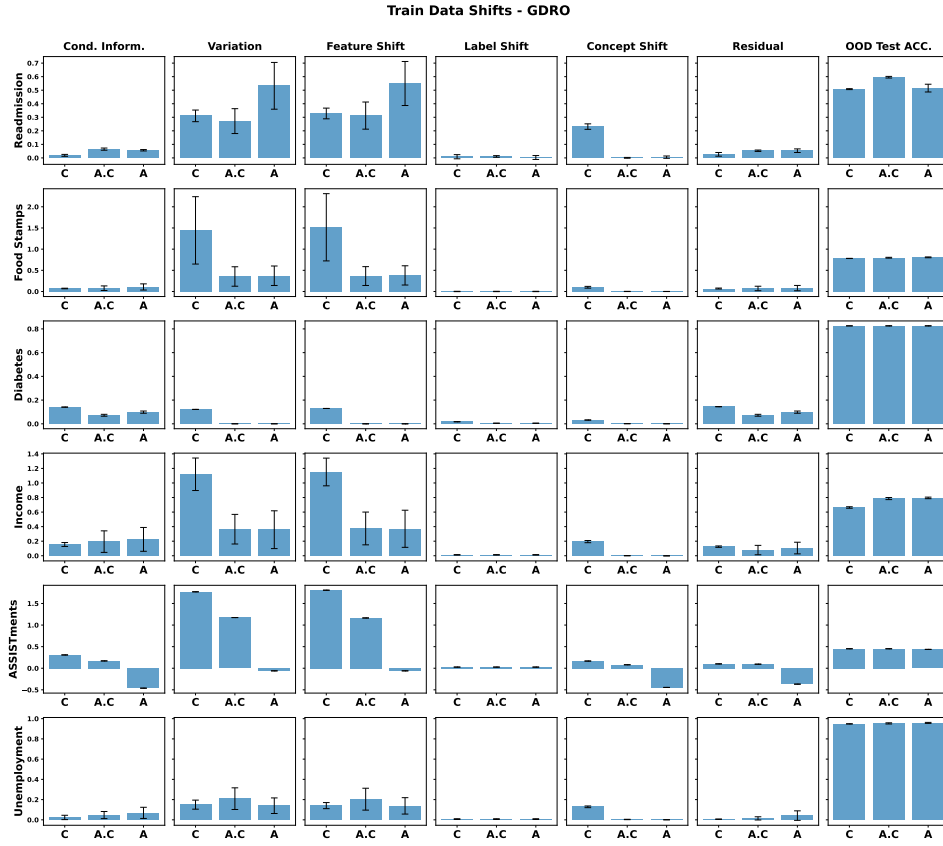


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

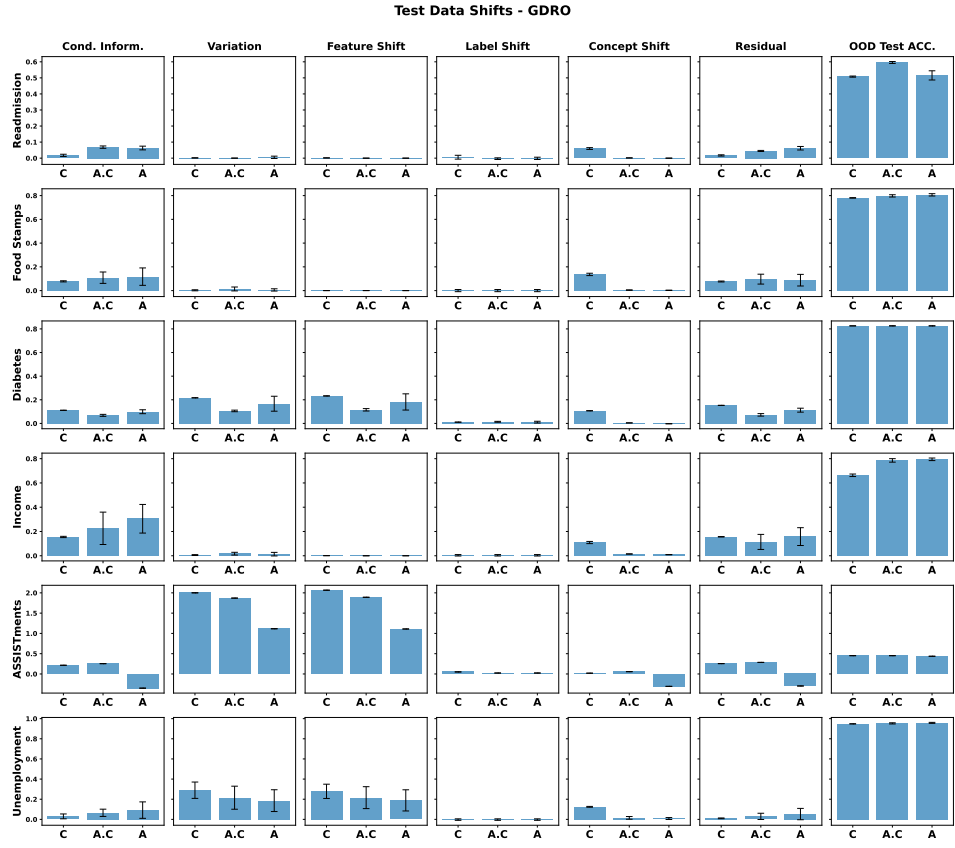


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

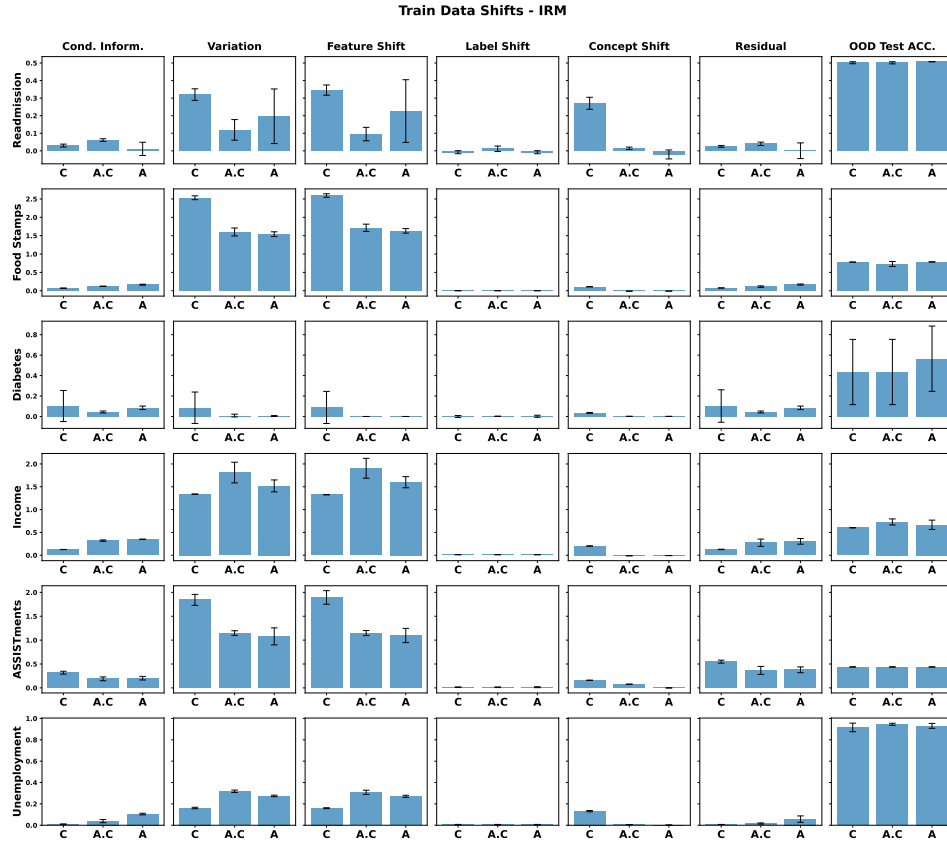


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

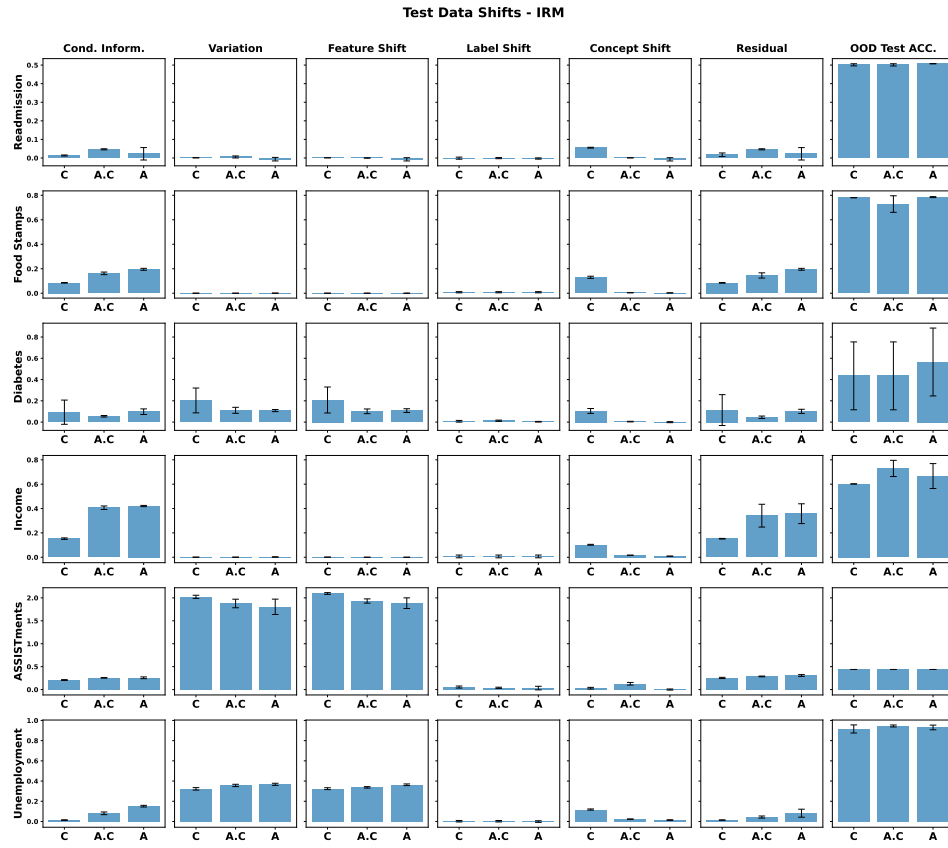


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

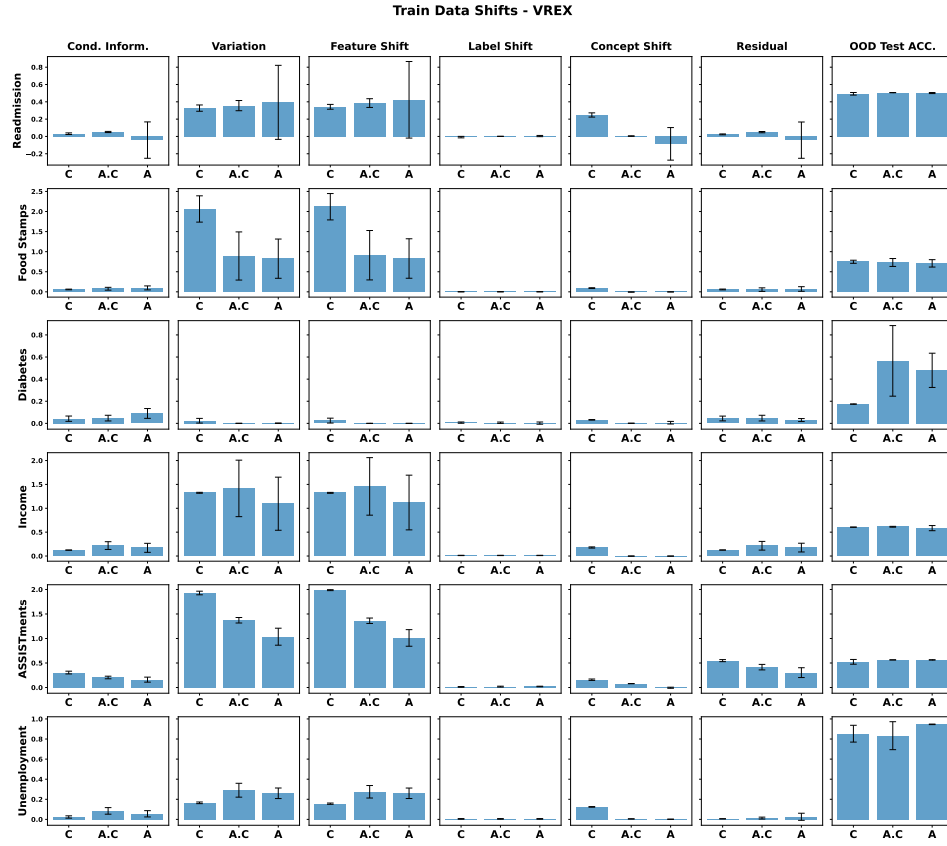


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

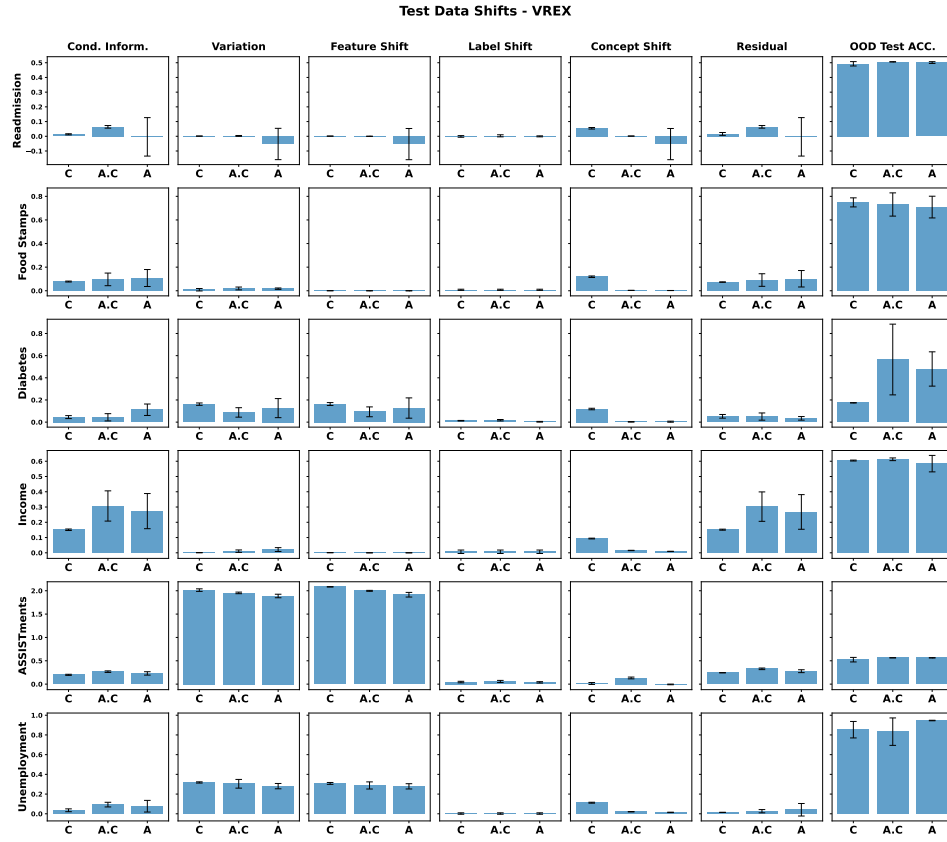


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

803 D Results on synthetic datasets

804 We use synthetic datasets to isolate three mechanisms: (i) the effect of alternative informativeness cri-
 805 teria, (ii) the dependence of model performance on confounder-support overlap across environments,
 806 and (iii) the effect of the causal-to-anti-causal covariate ratio. The data-generating process is defined
 807 by (12) with causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is unobserved, X
 808 is observed, Y is the target, and $\mathbf{X}_I$ denotes additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

809 We vary the functional forms of f_X, f_i , and f_Y across experiments. Domain shifts are generated by
 810 changing the environment-specific confounder mean μ_u^e , which controls the degree of distributional
 811 separation between environments and permits direct evaluation of model behavior under different
 812 shift magnitudes.

813 **Informativeness vs. accuracy:** We first use linear structural equations $f_X = 0.3U, f_i = 0.1U$, and
 814 $f_Y = X - 2U$, with $\mu_u^e \in \{-2, 2\}$ for training environments and $\mu_u^e \in \{0, 4\}$ for test environments.
 815 For Figure 5 in the main paper, we use one hidden confounder and $|\mathbf{X}_I| = 20$ informative covariates.
 816 Increasing the number of informative covariates from 0 to 20 reduces mean squared error, increases
 817 conditional informativeness, increases feature shift, and decreases concept shift. We then evaluate
 818 two additional informativeness regimes. First, with one hidden confounder and one informative
 819 covariate ($|\mathbf{X}_I| = 1$), we vary the proxy noise $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 shows that
 820 decreasing noise from 2.0 to 0.0 reduces mean squared error, increases conditional informativeness
 821 and feature shift, and decreases concept shift. As in the other synthetic experiments, the variation
 822 term remains zero. Second, we consider 20 hidden confounders with 20 corresponding informative
 823 covariates and modified structural equations $f_X = 0.3U, f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17
 824 shows the same pattern as informative covariates are introduced. These experiments specify multiple
 825 mechanisms by which informativeness improves OOD generalization.

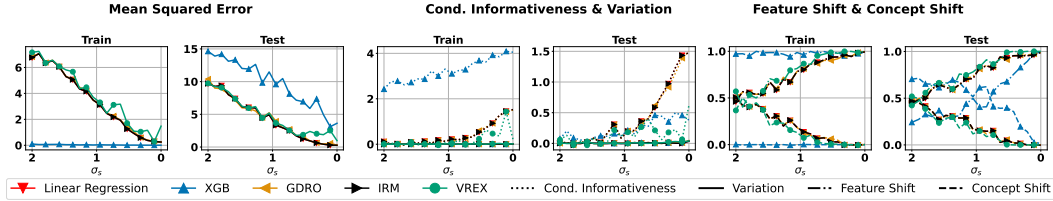


Figure D16: A low-noise proxy variable reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

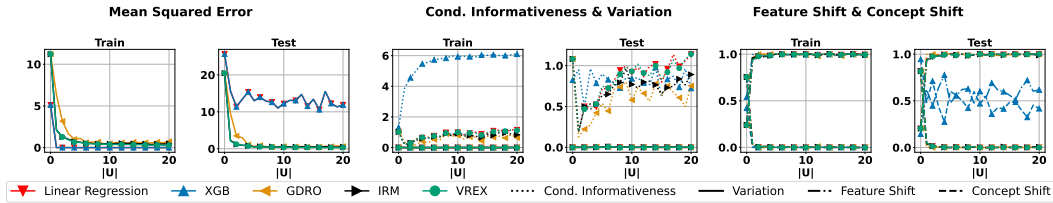


Figure D17: Adding proxy variables $\mathbf{X}_I$ for hidden confounders $\mathbf{U}$ that are informative about Y reduces MSE, increases conditional informativeness and feature shift, and decreases concept shift.

826 **Overlapping confounder support vs. accuracy:** As discussed in the main paper, recent work
 827 by Prashant et al. [62] proposes an OOD method for hidden confounding shift under the assumption
 828 that test confounder support is contained in training confounder support. We extend the linear
 829 synthetic analysis to nonlinear data with MLP and XGB models under both low- and high-overlap
 830 confounder-support regimes (Figures D18–D21). In the low-overlap regime, where ID and OOD
 831 values of μ_e are distant, models trained only on X learn an inadequate global function that does
 832 not separate environments. Adding environment-specific summary statistics of observed covariates

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

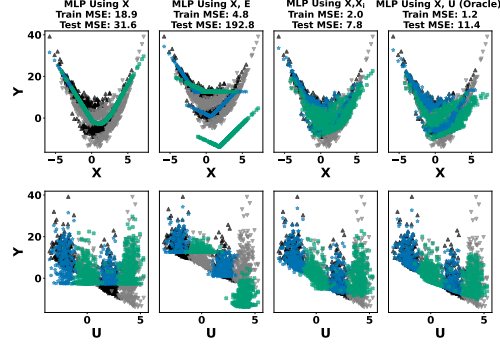


Figure D18: MLP performance in the low-overlap nonlinear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

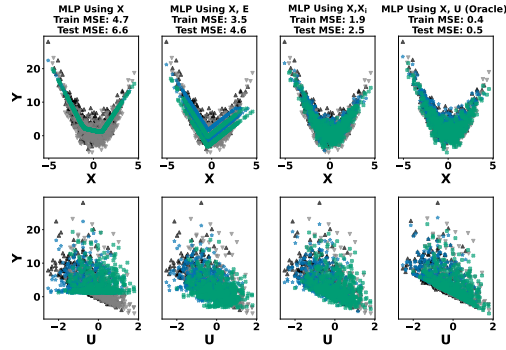


Figure D19: MLP performance in the high-overlap nonlinear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (oracle).

improves performance by encoding environment-specific relationships, but the largest gains occur when additional informative covariates are observed. We also include oracle results and U - Y scatter plots to clarify the learned input-output mappings. Performance is higher under high confounder overlap, as expected, but the low-overlap results are particularly relevant because they challenge the common-support assumption used in prior work.

Number of causal vs anti-causal variables: Recent benchmarks [24] report that baseline methods such as XGB and MLP often match or outperform OOD generalization methods such as GDRO, IRM, and VREX on real-world datasets, where the dominant tabular structure is $\mathbf{X} \rightarrow Y$. To test whether this behavior is explained by the causal structure, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$; for example, $\rho = 0.5$

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

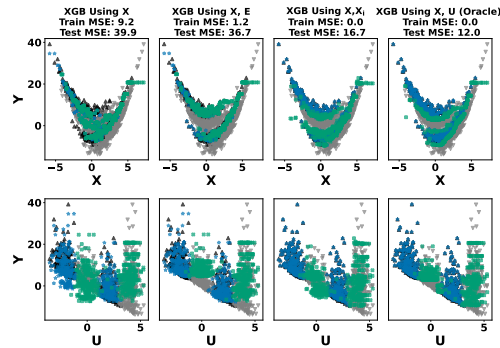


Figure D20: XGB performance in the low-overlap nonlinear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (oracle).

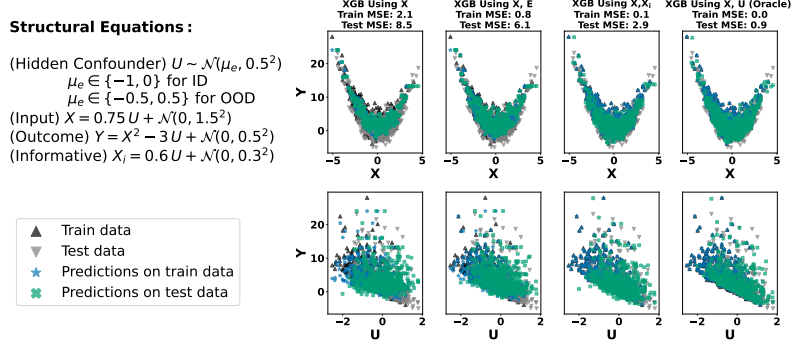


Figure D21: XGB performance in the high-overlap nonlinear setting for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (oracle).

gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$. Figure D22 shows that baselines remain competitive even when the causal structure is $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, although no method dominates for all settings (Figures D23, D24). Baselines achieve stronger PMA-OOD scores [24], defined as the fraction of maximum OOD accuracy. The evaluated methods are: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) robust optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

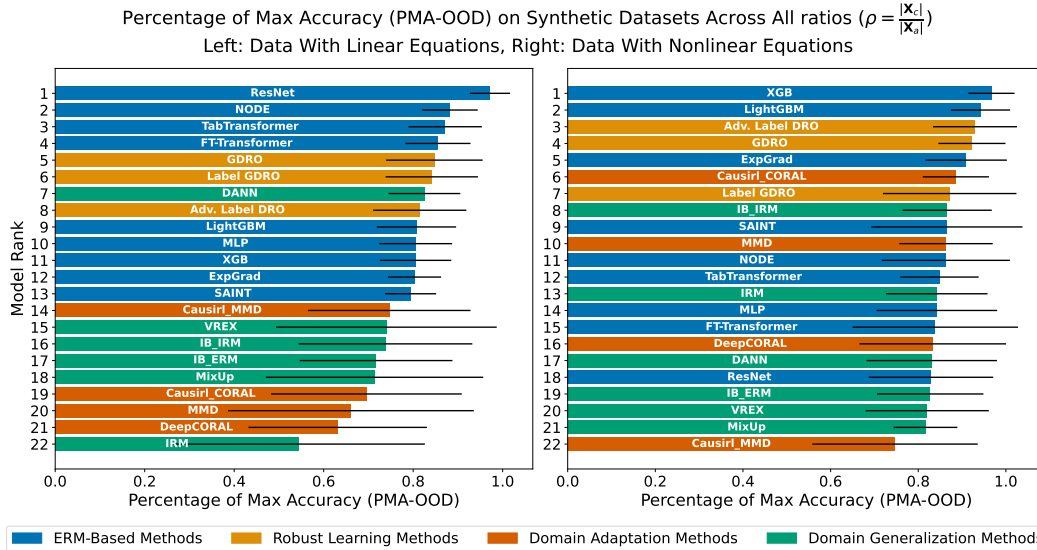


Figure D22: ERM-based methods achieve higher percentage-of-maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, consistent with the real-world results of [24].

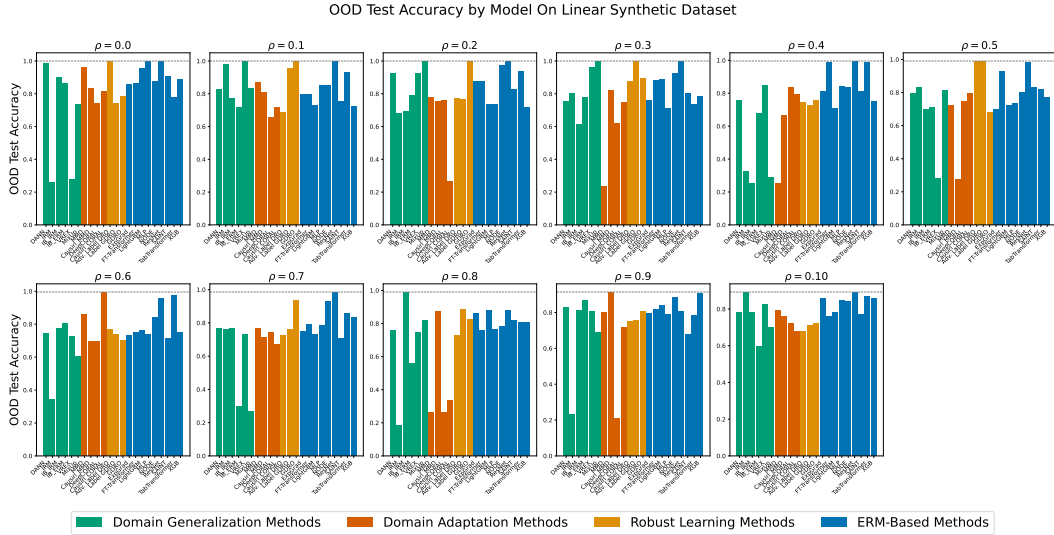


Figure D23: Model-specific OOD test accuracy in the linear setting as a function of ρ .

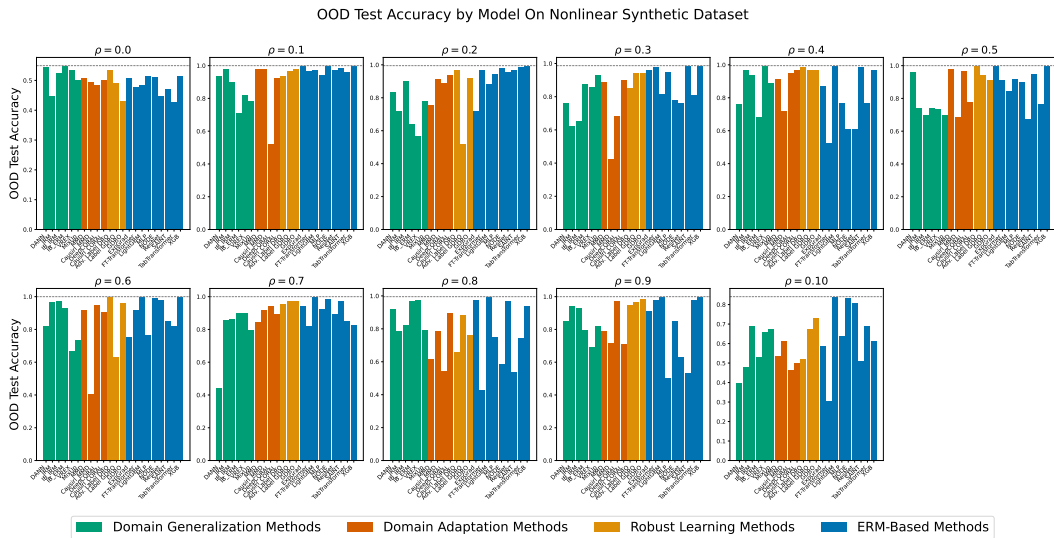


Figure D24: Model-specific OOD test accuracy in the nonlinear setting as a function of ρ .

E Qualitative analysis of hidden confounding shift in real-world datasets

This section qualitatively assesses hidden confounding shift in real-world datasets. Tables E3-E18 list plausible hidden confounders that may affect observed covariates across several datasets [57]. Dataset details are available in [24, 57]. We use GPT4o [1] to generate candidate confounders and treat the resulting lists as illustrative rather than exhaustive. The analysis provides domain-level support for the claim that hidden confounding shifts can occur across real-world prediction tasks.

Prompt: We use the following prompt to query GPT4o for candidate hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.